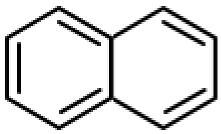
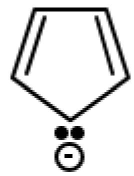
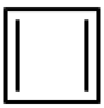
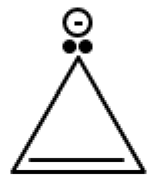
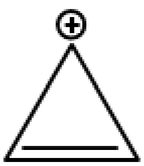
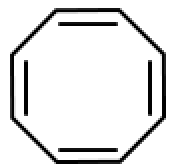
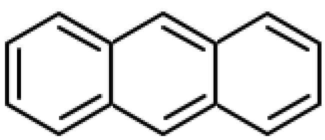
UPDATED SYLLABUS	HYDROCARBONS	60 MINUTES
SPECIALISED PYQ		40 QUESTIONS

Q1. Consider the following compounds/species: NEET - 2023

i.		ii.	
iii.		iv.	
v		vi	
vii			

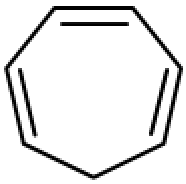
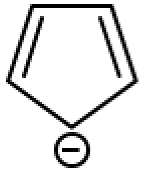
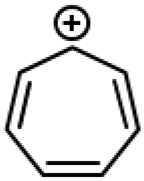
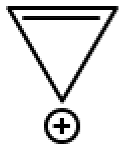
The number of compounds/species which obey Huckel's rule is:

1.	5	2.	4
3.	6	4.	2

Q2. Which amongst the following compounds will show geometrical isomerism?
NEET - 2023

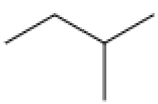
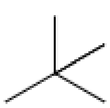
1. Pent-1-ene
2. 2,3-Dimethylbut-2-ene
3. 2-Methylprop-1-ene
4. 3,4-Dimethylhex-3-ene

Q3. Which of the following compounds does not exhibit aromaticity? NEET - 2022

1.		2.	
3.		4.	

Q4. Compound X on reaction with O_3O3 followed by $Zn/H_2OZn/H2O$ gives formaldehyde and 2-methyl propanal as products. The compound X is : NEET - 2022
1. Pent-2-ene
2. 3-Methylbut-1-ene
3. 2-Methylbut-1-ene
4. 2-Methylbut-2-ene

Q5. Match the bond line structures of hydrocarbons given in List I with the corresponding boiling points given in List II. NEET - 2022

	List-I (Compound)		List II (Boiling point in K)
(a)		(i)	300.9
(b)		(ii)	282.5
(c)		(iii)	309.1
(d)		(iv)	341.9

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(i)	(iv)	(iii)	(ii)

2.	(iii)	(i)	(iv)	(ii)
3.	(iii)	(iv)	(i)	(ii)
4.	(iv)	(i)	(ii)	(iii)

Q6. The compound obtained by addition of water to an alkyne having more than two carbons, in the presence of $HgSO_4$HgSO4 and dilute H_2SO_4H2SO4 at 333K, is: NEET - 2022

1.	A vicinal diol	2.	An aldehyde
3.	An alcohol	4.	A ketone

Q7. Given the following compounds: NEET - 2022

(a) Heptane
(b) Butane
(c) 2-Methylbutane
(d) 2-Methylpropane
(e) Hexane

For the compounds given above, the correct option capturing decreasing order of boiling points is:

1.	(a) > (c) > (e) > (d) > (b)	2.	(c) > (d) > (a) > (e) > (b)
3.	(a) > (e) > (b) > (c) > (d)	4.	(a) > (e) > (c) > (b) > (d)

Q8. The dihedral angle of the least stable conformer of ethane is: NEET - 2021

1.	60°	2.	0°
3.	120°	4.	180°

Q9. The major product of the following chemical reaction is: NEET - 2021

$$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>CH-CH=CH_2 + HBr \xrightarrow{(C_6H_5CO)_2O_2} ?$$

1.	$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>CH-\underset{\mid \atop Br}{CH}-CH_3$
2.	$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>CBr-CH_2-CH_3$
3.	$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>CH-CH_2-CH_2-Br$
4.	$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>CH-CH_2-CH_2-O-COC_6H_5$

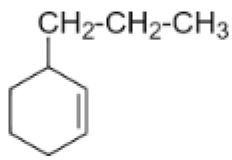
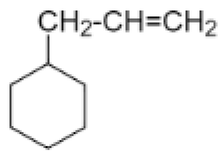
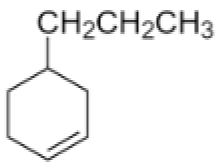
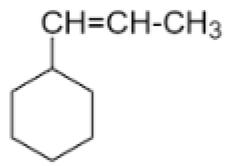
Q10. Which of the following statement(s) is/are correct about the elimination reaction of 2-Bromopentane to form pent-2-ene: NEET - 2020
(a) ββ-Elimination reaction
(b) Follows Zaitsev rule
(c) Dehydrohalogenation reaction
(d) Dehydration reaction

1.	(a), (c), (d)	2.	(b), (c), (d)
3.	(a), (b), (d)	4.	(a), (b), (c)

Q11. Which of the following alkane cannot be made in good yield by Wurtz reaction? NEET - 2020

1.	2,3-Dimethylbutane	2.	n-Heptane
3.	n-Butane	4.	n-Hexane

Q12. An alkene on ozonolysis gives methanal as one of the products. Its structure is: NEET - 2020

1.		2.	
3.		4.	

Q13. Which of the following is a free radical substitution reaction? NEET - 2020

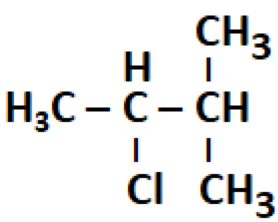
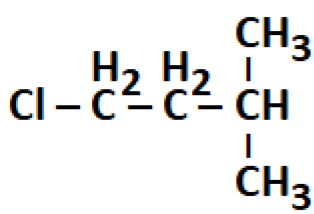
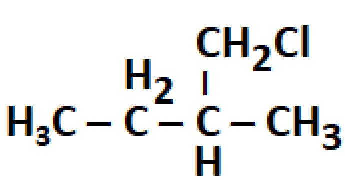
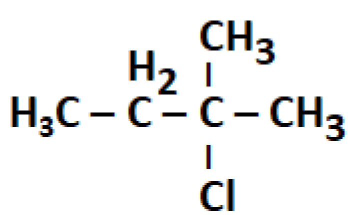
1. Benzene with $Br_2/AlCl_3$

2. Acetylene with HBr

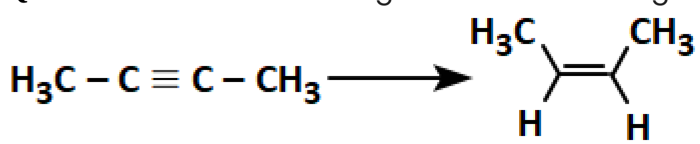
3. Methane with Br_2/hv

4. Propene with $HBr/(C_6H_5COO)_2$

Q14. An alkene "A" on reaction with O_3O3 and $Zn-H_2O$Zn-H2O gives propanone and ethanal in equimolar ratio. The addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is: NEET - 2019

1.		2.	
3.		4.	

Q15. The most suitable reagent for the following conversion is- NEET - 2019



cis-but-2-ene

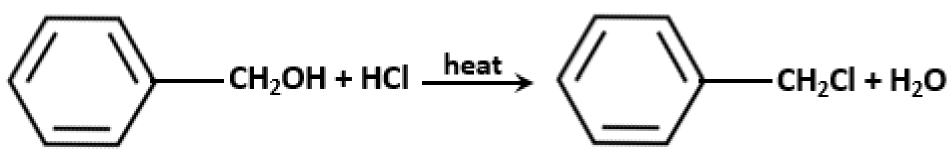
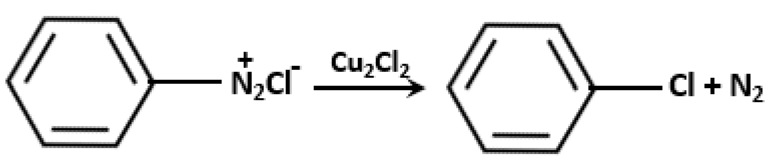
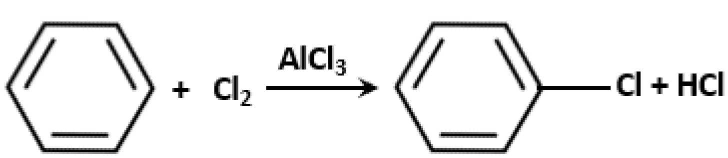
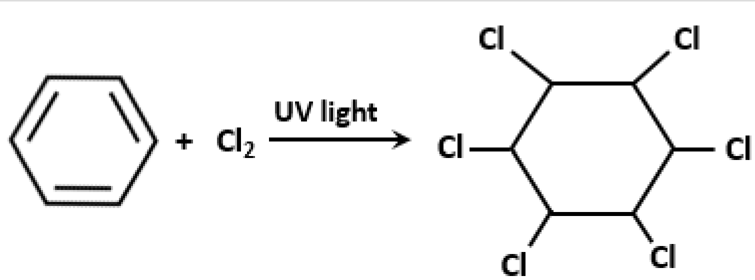
1. Hg^{2+}/ H^+, H_2O

2. Na/liquid NH_3

3. H_2, Pd/C, quinoline

4. Zn/HCl

Q16. The reaction among the following that proceeds through an electrophilic substitution reaction is: NEET - 2019

1.	
2.	
3.	
4.	

Q17. The alkane that gives only one monochloro product on chlorination with Cl_2Cl2 in presence of diffused sunlight is- NEET - 2019

1.	2,2,-dimethylbutane	2.	neopentane
3.	n-pentane	4.	Isopentane

Q18. In the following reaction, NEET - 2019

$CH_3C\equiv CH \xrightarrow[873K]{\text{Red hot iron tube}} A$CH3C≡CH→Red hot iron tube873K A

The number of ($\sigma\sigma$) bonds present in the product (A) is:

1. 21

2. 9

3. 24

4. 18

Q19. Hydrocarbon (A) reacts with bromine by substitution reaction to form an alkyl bromide B.
B undergoes the Wurtz reaction to give a gaseous hydrocarbon containing less than four carbon atoms. NEET – 2018

The formula of (A) is:

1. $CH\equiv CH$
2. $CH_2=CH_2$
3. CH_3-CH_3
4. CH_4

Q20. The compound C_7H_8C7H8 undergoes the following reactions NEET - 2018

$C_7H_8 \xrightarrow{3Cl_2/h\nu} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$C7H8→3Cl2/hvA→Br2/FeB→Zn/HCl C

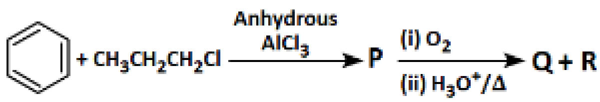
The product 'C' is-

1. m–Bromotoluene

2. o–Bromotoluene

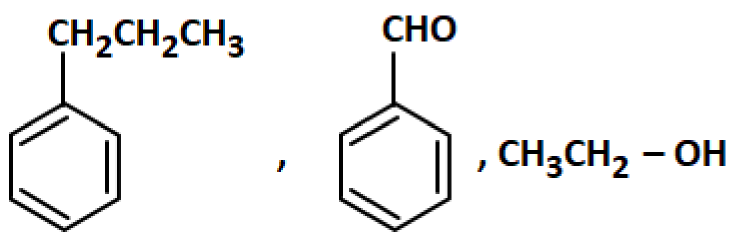
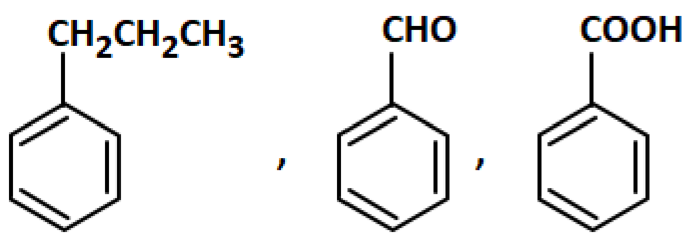
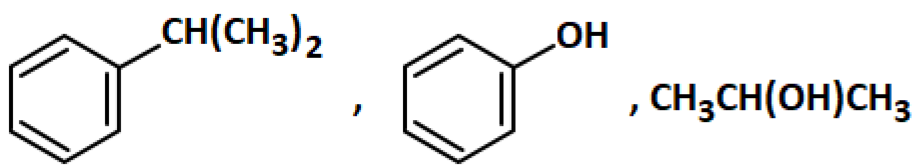
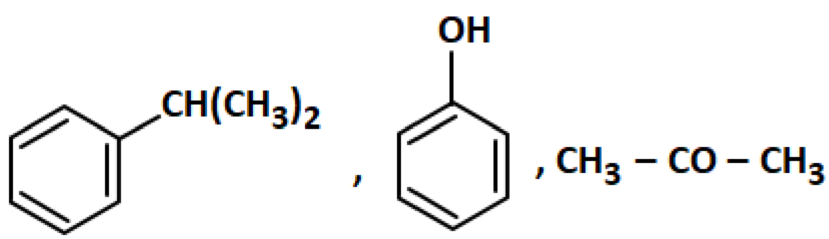
3. 3−Bromo−2,4,6−trichlorotoluene

4. p−Bromotoluene

Q21. 

NEET - 2018

P, Q and R in the above-mentioned sequence of reactions are respectively:

1.	
2.	
3.	
4.	

Q22. The molecule among the following that has the hybridization sp2, sp2, sp, and spsp2, sp2, sp, and sp from left to right atoms is: NEET - 2018

1. HC≡C−C≡HHC≡C−C≡H

2. $CH_2=CH-C\equiv CH$CH2=CH-C≡CH

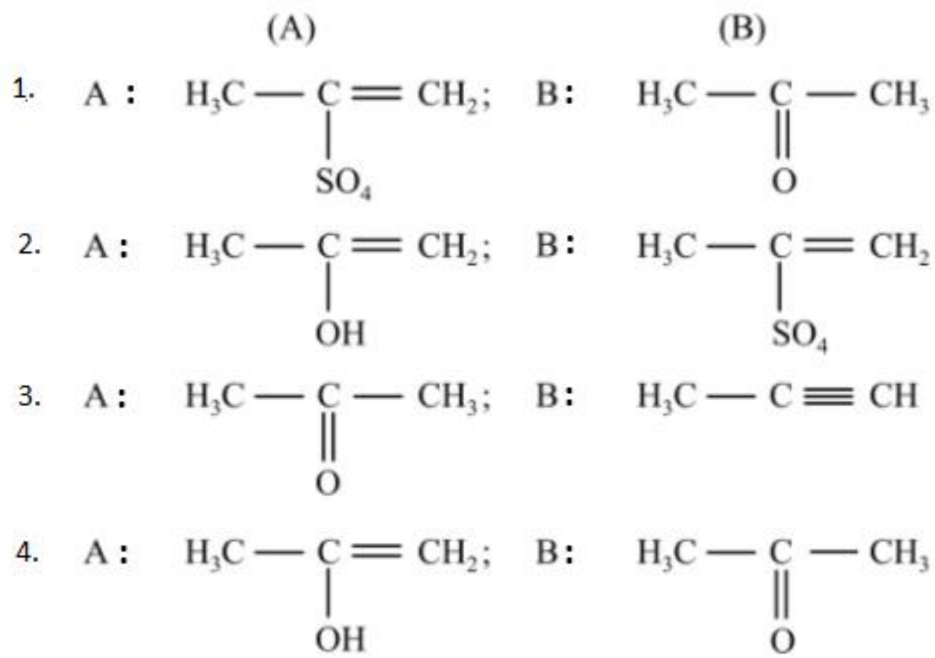
3. $CH_2=CH-CH=CH_2$CH2=CH-CH=CH2

4. $CH_2-CH=CH-CH_3$CH2-CH=CH-CH3

Q23. Predict the correct intermediate and product in the following reaction. NEET - 2017

$H_3C-C\equiv$≡CH H_2O,−⟶$HgSO_4$⟶HgSO4H2O, H_2SO_4H2SO4 Intermediate →⟶Product

(A) (B)



Q24. The correct order of acidity among the following is: NEET - 2017

1.	$CH\equiv CH>CH_3-C\equiv CH$CH≡CH>CH3−C≡CH $>CH_2=CH_2>CH_3-CH_3$>CH2=CH2>CH3−CH3
2.	$CH\equiv CH>CH_2=CH_2$CH≡CH>CH2=CH2 $>CH_3-C\equiv CH>CH_3-CH_3$>CH3−C≡CH>CH3−CH3
3.	$CH_3-CH_3>CH_2=CH_2$CH3−CH3>CH2=CH2 $>CH_3-C\equiv CH>CH\equiv CH$>CH3−C≡CH>CH≡CH

4.	$CH_2=CH_2>CH_3-CH_3$CH2=CH2>CH3−CH3 $>CH_3-C\equiv CH>CH\equiv CH$

Q25. Predict the correct intermediate and product in the following reaction: NEET - 2017

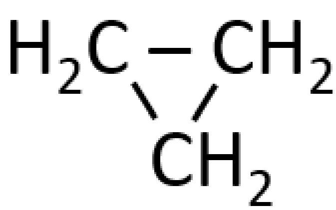
$$H_3C-C\equiv CH \xrightarrow[HgSO_4]{H_2SO_4\,,\,H_2O} \underset{A}{\text{Intermediate}} \longrightarrow \underset{\mathbf{B}}{\text{Product}}$$

1. A:A: $H_3C-C_{|OH}=CH_2$H3C−C|OH=CH2 ;; B:B: $H_3C-C_{|HSO_4}=CH_2$H3C−C|HSO4=CH2

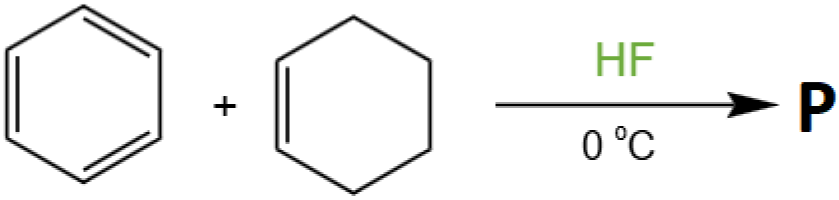
2. A:A: $H_3C-C_{||O}-CH_3$H3C−C||O−CH3 ;; B:B: $H_3C-C\equiv CH$H3C−C≡CH
3. A:A: $H_3C-_{OH|}C=CH_2$H3C−C|OH=CH2 ;; B:B: $H_3C-_{O||}C-CH_3$H3C−C||O−CH3

4. A:A: $H_3C-C_{|HSO_4}=CH_2$H3C−C|HSO4=CH2 ;; B:B: $H_3C-C_{||O}-CH_3$H3C−C||O−CH3

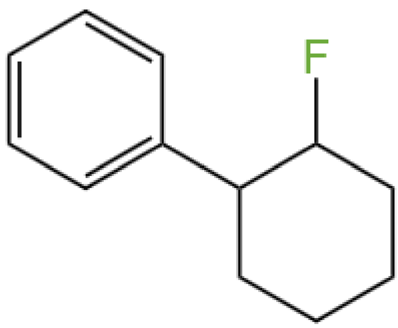
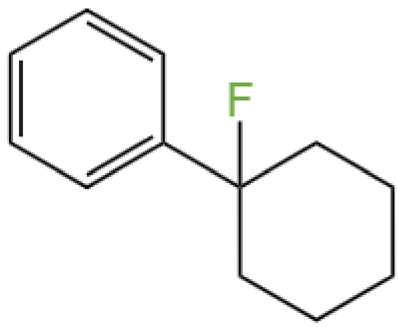
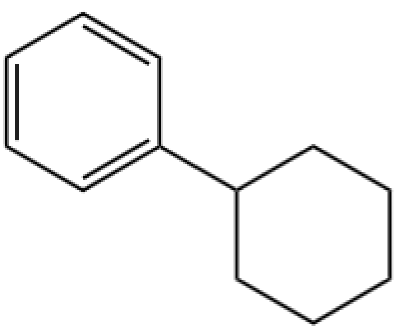
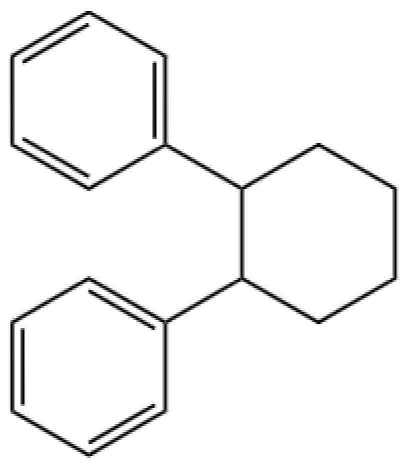
Q26. Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only elimination reaction? NEET - 2016

1.	$H_3C-CH_2-CH_2Br$
2.	
3.	$H_2C=C=O$
4.	$H_3C-CH_2-CH_2OH$

Q27. In the given reaction NEET - 2016



The product P is:

1.		2.	
3.		4.	

Q28. The compound that will react most readily with gaseous bromine has the formula:

NEET - 2016

1.	C_3H_6C3H6	2.	C_2H_2C2H2
3.	C_4H_{10}C4H10	4.	C_2H_4C2H4

Q29. The correct statement regarding the comparison of staggered and eclipsed conformations of ethane is: NEET - 2016

1.	The eclipsed conformation of ethane is more stable than staggered conformation because eclipsed conformation has no torsional strain.
2.	The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has a torsional strain.
3.	The staggered conformation of ethane is more stable than eclipsed conformation because staggered conformation has no torsional strain.
4.	The staggered conformation of ethane is less stable than eclipsed conformation because staggered conformation has a torsional strain.

Q30.

$$H-C\equiv CH \xrightarrow[\text{(ii) } CH_3CH_2Br]{\text{(i) } NaNH_2/liq.NH_3} X \xrightarrow[\text{(ii) } CH_3CH_2Br]{\text{(i) } NaNH_2/liq.NH_3} Y$$

X and Y in the above-mentioned reaction are respectively: NEET - 2016

1. X = 2–Butyne; Y = 3–Hexyne
2. X = 2–Butyne; Y = 2–Hexyne
3. X = 1–Butyne; Y = 2–Hexyne
4. X = 1–Butyne; Y = 3–Hexyne

ANSWER KEY

1. Ans. (2)
2. Ans. (4)
3. Ans. (1)
4. Ans. (2)
5. Ans. (3)
6. Ans. (4)
7. Ans. (4)
8. Ans. (2)
9. Ans. (3)
10. Ans. (4)
11. Ans. (2)
12. Ans. (2)
13. Ans. (3)
14. Ans. (4)
15. Ans. (3)
16. Ans. (3)
17. Ans. (2)
18. Ans. (1)
19. Ans. (4)
20. Ans. (1)
21. Ans. (4)
22. Ans. (2)
23. Ans. (4)
24. Ans. (1)
25. Ans. (3)
26. Ans. (3)
27. Ans. (3)
28. Ans. (1)
29. Ans. (3)
30. Ans. (4)